Novel mathematical machinery for VLSI macro placement

The macro placement problem — positioning N rectangular hard macros on a chip canvas — possesses extraordinary mathematical structure that remains almost entirely unexploited. Its feasible region is a union of exponentially many convex polyhedra (one per disjunctive assignment of pairwise left/right/above/below relations), and given any fixed assignment, the problem collapses to a linear program. This "LP inside combinatorics" architecture creates a bridge to at least seven deep mathematical frameworks — from tropical geometry to spin glass theory to optimal transport — none of which have been formally applied to placement. The Partcl/HRT Macro Placement Challenge 2026 ($49K+ in prizes, deadline May 21, 2026, GitHub ICCAD04 IBM benchmarks with 246–537 macros) provides an immediate testing ground. github GitHub Below is a synthesis of the most promising theoretical directions, ranked by the directness of their mathematical connection and the feasibility of producing working algorithms.

Tropical geometry speaks the placement problem's native language

The single most elegant mathematical connection is to tropical geometry. The half-perimeter wirelength for a net with pins at positions ${x_1, \ldots, x_k}$ is $\text{HPWL} = (\max_i x_i - \min_i x_i) + (\max_i y_i - \min_i y_i)$. arXiv In the $(\max, +)$ tropical semiring, this is **literally a tropical polynomial** — the placement objective lives natively in tropical algebraic geometry. The feasible polyhedra under each disjunctive assignment are classical polyhedra whose tropicalization yields tropical polyhedra, and the full feasible region forms a tropical polyhedral complex.

Three concrete developments make this actionable. First, Allamigeon, Gaubert, and Joswig showed that **the tropical central path characterizes the piecewise-linear limit of interior-point LP methods**, providing combinatorial lower bounds on LP complexity via the tropical geometry of the normal fan. SIAM +2 For placement, this means the combinatorial types of optimal placements are encoded in the secondary fan (normal fan of the state polytope) of the parametric LP as wirelength weights vary. Second, Talbut and Monod's "tropical gradient descent" (arXiv:2405.19551, 2024) achieves convergence rates matching classical gradient descent on problems with tropical convexity but not classical convexity arXiv — directly applicable since the HPWL objective is tropically convex. Third, Hashemi et al.'s "tropical attention" (arXiv:2505.17190, 2025) builds neural architectures in tropical projective space that universally approximate tropical circuits while preserving the sharp polyhedral decision boundaries that standard smooth activations destroy. arXiv

The theoretical gap requiring new work is a **tropical theory of disjunctive programming** — expressing the 4-term non-overlap disjunction in tropical semiring operations and characterizing the tropical variety of optimal solutions as cost vectors vary. Current tropical LP algorithms have exponential internal representations in high dimensions, Hal so scaling to the $2N$-dimensional placement space (with $N \sim 500$ macros) demands exploiting the sparse, factored structure of pairwise constraints.

The cavity method maps placement to a solvable spin system

The placement problem maps naturally onto a factor graph where each macro pair (i,j) carries a **4-state Potts-like spin** $\sigma_{ij} \in \{L, R, A, B\}$ encoding relative position. The partition function integrates over all spin configurations weighted by the LP-optimal cost for each configuration's induced polyhedron. This is precisely the structure where the cavity method and survey propagation (Mézard, Parisi, Zecchina — *Science*, 2002) have produced algorithmic breakthroughs.

Survey propagation computes distributions over the clustered solution space under **replica symmetry breaking (RSB)**. Springer The 1RSB cavity method would identify clusters of disjunctive assignments that are internally connected (nearby in Hamming distance on the σ_{ij}'s), characterize how these clusters relate hierarchically, and guide search toward under-constrained variables. The union-of-polyhedra structure maps directly: each polyhedron is one spin configuration, and clusters of nearby polyhedra correspond to the RSB cluster hierarchy. The phase transition phenomenology from random CSP theory (Krzakala, Montanari et al., *PNAS*, 2007) predicts that as the ratio of total macro area to canvas area increases, the solution space undergoes a cascade: single connected cluster → exponentially many clusters (clustering transition) → condensation into few dominant clusters → infeasibility threshold. **For the ICCAD04 benchmarks at 43–53% area utilization, the problem likely sits near or beyond the clustering transition**, GitHub explaining why local search gets trapped.

The critical theoretical development needed is **hybrid discrete-continuous belief propagation** — message-passing that marginalizes over LP optima. Each factor-to-variable message must encode the parametric LP optimal value as a function of adjacent variable assignments. The spatial embedding creates short loops (nearby macro triples have correlated constraints), violating the locally-tree-like assumption. Region-graph methods (Kikuchi approximations) or loop-corrected BP would be necessary. Springer No application of cavity methods to geometric placement with disjunctive constraints has been published.

Complexification dissolves the disconnectedness barrier

The fundamental obstacle in macro placement optimization is that the real feasible region is disconnected — local search within one polyhedron cannot reach other polyhedra without crossing infeasible (overlapping) regions. **Complexification eliminates this barrier entirely.** When coordinates are extended from $\mathbb{R}^{2N}$ to $\mathbb{C}^{2N}$, the hyperplane "walls" between polyhedra (where $x_j - x_i - w_i = 0$) become complex hyperplanes. A classical result from arrangement theory states that the complement of a complex hyperplane arrangement in $\mathbb{C}^n$ is always path-connected. Therefore, **paths exist in $\mathbb{C}^{2N}$ connecting any two feasible placement configurations while avoiding all constraint boundaries**.

The algorithmic realization uses numerical algebraic geometry (Sommese and Wampler; ResearchGate the Bertini software). Springer Reformulate non-overlap constraints using complementarity: introduce slack variables $s_{ij,k} \geq 0$ for each of the four disjuncts per pair, with the product constraint $\prod_k s_{ij,k} = 0$ (at least one must be active). In $\mathbb{C}^N$, homotopy continuation can trace paths between optima of different polyhedra through complex space, then project back to real solutions. Parameter homotopies — tracking how optima move as macro sizes, canvas dimensions, or wirelength weights vary — would reveal when and how local minima merge, split, or disappear, Mountainscholar connecting directly to catastrophe-theoretic bifurcations. The Gaussian homotopy approach (Mobahi and Fisher) provides a complementary strategy: convolving the non-convex objective with a Gaussian kernel of decreasing bandwidth creates a smooth path from a convex approximation to the original function, University of Maryland Department effectively "filling in" the infeasible gaps between polyhedra.

Optimal transport reframes density as a natural geometric flow

The density minimization component of placement is **fundamentally an optimal transport problem**. Given N macros with total area A distributed with some density μ_0, the target is a uniform density $\mu_{\text{target}} = A / |\text{Canvas}|$. The state-of-the-art ePlace/DREAMPlace framework already performs approximate OT implicitly — its electrostatic potential ϕ solving the Poisson equation is the Brenier transport potential where the optimal map is $T(x) = x - \nabla\phi(x)$. Cloudfront

Making this connection explicit unlocks three advances. First, **entropy-regularized optimal transport via Sinkhorn iterations** (Cuturi, NeurIPS 2013) provides a differentiable, efficiently computable density-spreading mechanism with controllable regularization strength, replacing the ad-hoc smoothing in DREAMPlace. Second, the **Benamou-Brenier formulation** interprets global placement as a Wasserstein gradient flow: the placement evolves along the geodesic in Wasserstein space from concentrated initial positions toward uniform density, with wirelength acting as a potential energy. This gives a principled dynamics replacing the current heuristic annealing of density weights. Third, **semi-discrete optimal transport** (discrete point masses transported to continuous target) naturally produces Laguerre tessellations (power diagrams) that partition the canvas into macro "territories" — a placement-aware Voronoi decomposition that directly encodes congestion information.

The key missing piece is **constrained OT preserving rectangular geometry and non-overlap**. Standard OT does not enforce that transported regions remain axis-aligned rectangles. Developing this "geometric OT" for placement would require integrating the disjunctive structure into the transport framework — potentially via multi-marginal OT formulations where each net's routing demand is a separate marginal.

Stratified Morse theory characterizes the optimization landscape

The feasible region $F = \bigcup_\sigma P_\sigma$ is a union of polyhedra with shared faces — precisely a **stratified space** in the sense of Goresky and MacPherson's stratified Morse theory Springer (1988). The strata are: interiors of full-dimensional polyhedra (top strata), shared codimension-1 faces (where one disjunctive assignment changes), shared codimension-2 faces, and so on. Morse theory on this stratified space decomposes the local Morse data at each critical point into tangential data (within the stratum) and normal data (transverse to it). Intlpress Ias

On each polyhedron P_σ, the HPWL objective is piecewise-linear, so the LP optimum sits at a vertex or face — giving at most one critical point per polyhedron. **The Morse inequalities then bound the total number of local minima below by the Betti numbers of F.** Since F is a union of chambers in a hyperplane arrangement, its Betti numbers are computable from the intersection lattice via the Orlik-Solomon algebra and the Möbius function. Emergent Mind The Salvetti complex provides a finite CW-complex homotopy equivalent to the complexified arrangement complement, ResearchGate arXiv with cells corresponding to valid disjunctive assignments.

For algorithmic applications, **persistent homology of the objective filtration** $F_c = \{x \in F : \text{Objective}(x) \leq c\}$ reveals at which cost levels connected components (H_0), tunnels (H_1), and voids (H_2) appear and disappear. Long-lived H_0 features indicate well-separated basins of attraction for local search; short-lived features indicate noise. Ista This topological fingerprint could guide multi-start strategies by identifying the number and relative depth of distinct solution clusters before expensive optimization begins. Wikipedia

Disjunctive programming hierarchies provide certified bounds

The most mature applicable optimization theory comes from Balas's disjunctive programming. Wiley Online Library For the macro placement non-overlap constraints — each a 4-term linear disjunction — the convex hull of each pairwise disjunction can be computed in the extended variable space. A recent result (Springer LNCS, 2024) proves that the **"full optimal big-M lifting" convex hull formulation is exact for axis-aligned rectangles when $d \leq 2$**, Springer directly matching the placement geometry.

The P-split formulation hierarchy (Kronqvist et al., *Mathematical Programming*, 2025) provides a principled middle ground between the weak Big-M relaxation and the expensive convex hull formulation. At level $P=1$, one recovers Big-M; at the top level, one recovers the convex hull. Computational results on 344 instances show that intermediate P-split levels achieve node counts comparable to the full convex hull while reducing solution time by an order of magnitude. For placement, this means one can **progressively tighten the LP relaxation** by applying sequential convexification to the most constraining macro pairs first.

The work on ideal formulations for rectangle packing (arXiv:2601.15252) develops automated methods to prove that certain MBLP formulations project exactly onto the convex hull — the strongest possible relaxation. arXiv The **SDP relaxation approach of Anjos and Liers** (2012) provides global bounds on VLSI floorplanning SciSpace for up to ~100 facilities using mixed-integer SOCP-SDP formulations. Combining these with the Lasserre hierarchy's guaranteed convergence arXiv arXiv (O(1/r) rate, finite convergence generically per Nie 2014) and sparse variants like TSSOS that exploit the pairwise constraint structure could yield certifiably optimal solutions for small-to-medium instances.

Diffusion models and information geometry offer practical pathways

Two rapidly developing areas bridge deep theory with implementability. **Diffusion models for placement** have already produced working systems: Lee et al. (ICML 2025) replace sequential RL with simultaneous denoising of all macro positions, trained entirely on synthetic data, achieving competitive HPWL arXiv arXiv and **>35% congestion reduction** over prior methods. arXiv DiffPlace (arXiv:2510.15897) adds constrained manifold diffusion to enforce non-overlap during the denoising process. arXiv The DiffUCO framework (ICML 2024) applies diffusion to unsupervised combinatorial optimization using reverse KL divergence bounds, achieving state-of-the-art on MIS, MaxCut, and covering problems. arXiv +2 The theoretical gap is substantial — no approximation guarantees exist — but the empirical results suggest diffusion processes can effectively sample from the multimodal union-of-polytopes landscape.

Information geometry via the Fisher-Rao metric provides a principled optimization framework for the hybrid discrete-continuous structure. Müller and Montúfar et al.

(arXiv:2403.19448, 2024) proved that Fisher-Rao gradient flows achieve **linear convergence for linear programs**, with rate depending on LP geometry. arXiv For placement, parameterizing the probability distribution over disjunctive assignments as a product distribution $p(\sigma_{ij} = k \mid \theta)$ for $k \in {L,R,A,B}$ gives a natural exponential family. The Fisher information matrix is block-diagonal ($N(N-1)/2$ blocks of 3×3), making the natural gradient $\tilde{\nabla}|*theta = F(|theta)^{-1} |nabla*\theta \mathbb{E}[\text{cost}]$ computationally trivial. Emergent Mind Adding KL regularization $\text{KL}(p | q)$ yields a Boltzmann distribution $p^*(\sigma) \propto q(\sigma) \exp(-\beta \cdot \text{LP-cost}(\sigma))$ — reconnecting to the statistical physics picture and providing a principled annealing schedule.

The competition context and what approaches are viable

The Partcl/HRT Macro Placement Challenge 2026 evaluates on 17 ICCAD04 IBM benchmarks (246–537 macros, 7K–16K nets, **43–53% area utilization**) with proxy cost $1.0 \times \text{WL} + 0.5 \times \text{Density} + 0.5 \times \text{Congestion}$, subject to a 1-hour runtime on an AMD EPYC 9655P with NVIDIA RTX 6000 Ada 48GB. GitHub The RePlAce analytical placer baseline dominates SA across all benchmarks (15–55% lower proxy cost). Top-7 submissions undergo full OpenROAD flow evaluation on NG45 designs. GitHub github

The sequence pair representation (Murata et al., 1996) resolves the enumeration problem by construction — each pair of permutations (Γ^+, Γ^-) maps to a feasible placement, giving $(n!)^2$ elements University of Michigan rather than $4^{n(n-1)/2}$ raw assignments. ACM Digital Library IEEE Xplore The adjacency graph on sequence pairs (via adjacent transpositions) is connected with diameter $\Theta(n^2)$, ensuring reachability. The practical path forward combines these compact representations with the theoretical machinery above: use tropical gradient descent or Fisher-Rao flows for the continuous optimization within each polyhedron, survey propagation or diffusion for navigating between polyhedra, and disjunctive programming bounds for certification.

Conclusion

The macro placement problem sits at a remarkable intersection of mathematical structures — tropical, statistical-mechanical, topological, transport-theoretic — that have developed independently but converge on this single problem. **Three directions stand out for immediate theoretical impact.** First, tropical geometry provides the algebraically natural language for HPWL optimization, with tropical gradient descent already demonstrating practical convergence. Second, complexification via numerical algebraic geometry offers the only known principled method for connecting the disconnected real feasible regions, Springer Springer transforming a fundamentally discrete navigation problem into a continuous one. Third, the Fisher-Rao gradient flow for LPs combined with natural gradient updates on assignment distributions provides a complete, mathematically grounded optimization framework for the hybrid discrete-continuous structure. SIAM arXiv The deeper frameworks — cavity methods, persistent homology, random matrix theory — offer structural insights (phase transition locations, landscape complexity bounds, topological obstruction counts) that could guide algorithm design even before they yield direct algorithms. What distinguishes all these approaches from standard heuristics is that they engage with the problem's specific mathematical structure — the union-of-polyhedra geometry, the tropical objective, the Potts-like spin system — rather than treating it as a generic black-box optimization.